Bob's Report: Evaluating Store 10's Weekly Sales Performance

Consulting Case – Final Deliverable

> "*I had the data, but what was I supposed to do now?*" – Bob Garza

This notebook takes Bob's wrangled supermarket dataset and turns it into a defensible, model-based evaluation of **Store 10**'s weekly sales performance.

The report is structured around four clearly identified areas:

1. **Business question & data** – what we are trying to answer, and what we have to work with
2. **Modeling strategy** – two meaningfully different models and the logic behind comparing them
3. **Model comparison** – performance metrics and model selection, with justification
4. **Interpretation & recommendations** – what the best model tells us about Store 10, and what to do about it

---

1. Business Question and Data

1.1 The business question

Bob's first attempt concluded that Store 10 is a "*good performing store*" because its average weekly sales are higher than the dataset-wide average. That comparison is misleading: Store 10 is a **mid-sized Type B store**, and the dataset average mixes in very small Type C stores. Comparing apples to oranges makes any store with above-average size look like an over-performer.

We therefore reframe the question into something *measurable and decision-relevant*:

> **Given Store 10's characteristics (size, type, location-driven conditions) and the week being evaluated (holiday, season, macro environment), how do its actual sales compare to the sales we would *expect* from a similar store under similar conditions?**

Translated into an analytical task: we build a model that predicts the weekly sales a "peer-equivalent" store would produce, and we evaluate Store 10 by looking at the **gap between actual and predicted sales (the residual)**.

This turns a vague qualitative judgement into a **quantifiable over- or under-performance signal** that an executive can act on.

1.2 The data

The analysis uses the three Walmart-style datasets Bob received (already cleaned and joined in a prior wrangling step). We work at the **store-week level** (one row per store per week, 45 stores × ~143 weeks = 6,435 rows). Department-level granularity adds complexity without new signal for this question, so we aggregate sales up to the store-week.

Variable	Source	Description
`Store`	stores	Store identifier (1–45)
`Date`	sales	Week ending date
`Weekly_Sales`	sales (summed over depts)	**Target.** Total sales for the store that week (USD)
`IsHoliday`	sales	Whether the week contains a special holiday
`Type`	stores	Store type: A, B, or C (a proxy for format/assortment)

`Size`	stores	Store size in square feet
`Temperature`	features	Average weekly temperature in the store's region
`Fuel_Price`	features	Weekly fuel price in the store's region
`CPI`	features	Regional consumer price index
`Unemployment`	features	Regional unemployment rate

We will also engineer `Month` and `Week` (ISO week number) from `Date` to let the models learn seasonality (which matters a lot — Thanksgiving and Christmas weeks drive huge spikes).

1.3 Load and describe the data

In [1]:
```
# ----- Imports -----
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Display settings
pd.set_option("display.max_columns", 40)
pd.set_option("display.float_format", lambda x: f"{x:,.2f}")
sns.set_style("whitegrid")
plt.rcParams["figure.dpi"] = 100

RANDOM_STATE = 42
```

In [2]:
```
# ----- Load the cleaned store-week panel and merge store Type -----
df      = pd.read_csv("clean_wrangled_store_week.csv")
stores  = pd.read_csv("stores.csv")
df      = df.merge(stores[["Store", "Type"]], on="Store", how="left")

# Parse dates and add seasonal features
df["Date"]  = pd.to_datetime(df["Date"])
df["Year"]  = df["Date"].dt.year
df["Month"] = df["Date"].dt.month
df["Week"]  = df["Date"].dt.isocalendar().week.astype(int)

print(f"Rows: {len(df):,}   Stores: {df['Store'].nunique()}   "
      f"Weeks: {df['Date'].nunique()}")
print(f"Date range: {df['Date'].min().date()}  to  {df['Date'].max().date()}")
print(f"Missing values: {df.isna().sum().sum()}")

df.head()
```

```
Rows: 6,435   Stores: 45   Weeks: 143
Date range: 2010-02-05  to  2012-10-26
Missing values: 0
```

Out[2]:

	Store	Date	Weekly_Sales	IsHoliday	Temperature	Fuel_Price	CPI	Unemployment	Size	Typ
0	1	2010-02-05	1,643,690.90	0	42.31	2.57	211.10	8.11	151315	
1	1	2010-02-12	1,641,957.44	1	38.51	2.55	211.24	8.11	151315	
2	1	2010-02-19	1,611,968.17	0	39.93	2.51	211.29	8.11	151315	
3	1	2010-02-26	1,409,727.59	0	46.63	2.56	211.32	8.11	151315	
4	1	2010-03-05	1,554,806.68	0	46.50	2.62	211.35	8.11	151315	

In [3]:

```
# ----- Summary statistics -----
numeric_cols = ["Weekly_Sales","Size","Temperature","Fuel_Price","CPI","Unemployment"]
df[numeric_cols].describe().T
```

Out[3]:

	count	mean	std	min	25%	50%	75%	
Weekly_Sales	6,435.00	1,046,964.88	564,366.62	209,986.25	553,350.10	960,746.04	1,420,158.66	3,818,
Size	6,435.00	130,287.60	63,117.02	34,875.00	70,713.00	126,512.00	202,307.00	219,
Temperature	6,435.00	60.66	18.44	-2.06	47.46	62.67	74.94	
Fuel_Price	6,435.00	3.36	0.46	2.47	2.93	3.44	3.73	
CPI	6,435.00	171.58	39.36	126.06	131.74	182.62	212.74	
Unemployment	6,435.00	8.00	1.88	3.88	6.89	7.87	8.62	

1.4 Store 10 in context — why the naive comparison is misleading

Bob's original framing was *Store 10's average weekly sales exceed the dataset average, therefore Store 10 performs well*. Let's look at that comparison in its proper context before going further.

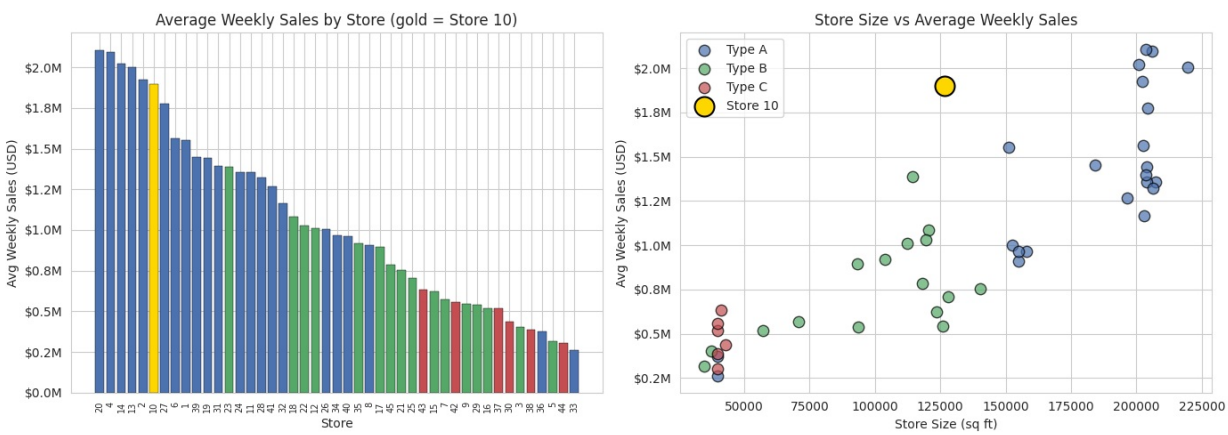
In [4]:

```
# ----- Naive vs peer-adjusted comparison -----
store_avg = (df.groupby(["Store", "Type"], as_index=False)
               .agg(Avg_Weekly_Sales=("Weekly_Sales", "mean"),
                    Size=("Size", "first")))

s10_row = store_avg[store_avg["Store"] == 10].iloc[0]

print("=== Naive comparison (what Bob originally did) ===")
print(f"Store 10 avg weekly sales:          ${s10_row['Avg_Weekly_Sales']:>14,.0f}")
print(f"All stores avg weekly sales:        ${store_avg['Avg_Weekly_Sales'].mean():>14,.0f}")
print(f"Difference:                         {(s10_row['Avg_Weekly_Sales'] / store_avg['Avg_Weekly_Sal

print("\n=== Peer-adjusted (same Type) ===")
same_type = store_avg[store_avg["Type"] == s10_row["Type"]]
print(f"Store 10 avg weekly sales:          ${s10_row['Avg_Weekly_Sales']:>14,.0f}")
print(f"Type-{s10_row['Type']} peer avg weekly sales:      ${same_type['Avg_Weekly_Sales'].mean():>14
print(f"Difference vs Type-{s10_row['Type']} peers:         {(s10_row['Avg_Weekly_Sales'] / same_type

print("\nStore 10 characteristics:")
print(f"  Type: {s10_row['Type']}   Size: {s10_row['Size']:,} sq ft")
```

```
=== Naive comparison (what Bob originally did) ===
Store 10 avg weekly sales:          $     1,899,425
All stores avg weekly sales:        $     1,046,965
Difference:                                  +81.4%

=== Peer-adjusted (same Type) ===
Store 10 avg weekly sales:          $     1,899,425
Type-B peer avg weekly sales:      $       822,995
Difference vs Type-B peers:                 +130.8%

Store 10 characteristics:
  Type: B   Size: 126,512 sq ft
```

In [5]:

```
# ----- Visual: average weekly sales by store, colored by Type, highlight Store 10 -----
fig, ax = plt.subplots(1, 2, figsize=(14, 5))

# Left: bar chart sorted by sales
bar_data = store_avg.sort_values("Avg_Weekly_Sales", ascending=False).reset_index(drop=True)
type_colors = {"A": "#4C72B0", "B": "#55A868", "C": "#C44E52"}
colors = [type_colors[t] for t in bar_data["Type"]]
# Highlight store 10
s10_idx = bar_data.index[bar_data["Store"] == 10][0]
colors[s10_idx] = "gold"
ax[0].bar(bar_data["Store"].astype(str), bar_data["Avg_Weekly_Sales"], color=colors, edgecolor="black
ax[0].set_title("Average Weekly Sales by Store (gold = Store 10)")
ax[0].set_xlabel("Store")
ax[0].set_ylabel("Avg Weekly Sales (USD)")
ax[0].tick_params(axis="x", labelsize=7, rotation=90)
ax[0].yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f"${x/1e6:.1f}M"))

# Right: size vs sales scatter
for t, g in store_avg.groupby("Type"):
    ax[1].scatter(g["Size"], g["Avg_Weekly_Sales"], label=f"Type {t}",
                  color=type_colors[t], s=80, alpha=0.7, edgecolor="black")
ax[1].scatter(s10_row["Size"], s10_row["Avg_Weekly_Sales"],
              color="gold", s=250, edgecolor="black", linewidth=1.5, label="Store 10", zorder=5)
ax[1].set_title("Store Size vs Average Weekly Sales")
ax[1].set_xlabel("Store Size (sq ft)")
ax[1].set_ylabel("Avg Weekly Sales (USD)")
ax[1].yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f"${x/1e6:.1f}M"))
ax[1].legend()

plt.tight_layout()
plt.show()
```

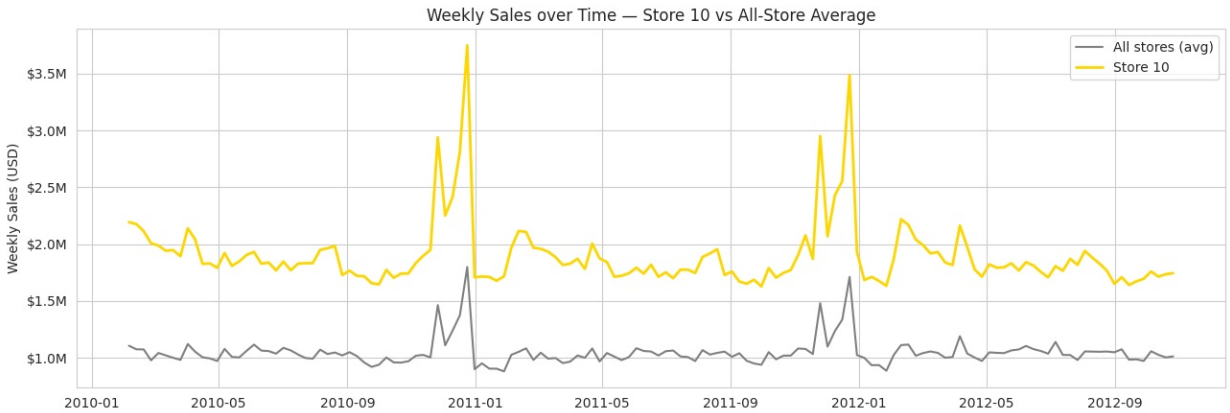


Read-out: Even the same-type comparison is unfair: Store 10 is on the *large* end of the Type B cluster (~127K sq ft vs a Type B average closer to ~100K sq ft), so its sales advantage over the Type B average is partly just a size effect. To evaluate Store 10 fairly we need a model that simultaneously controls for size, type, seasonality, and macro conditions.

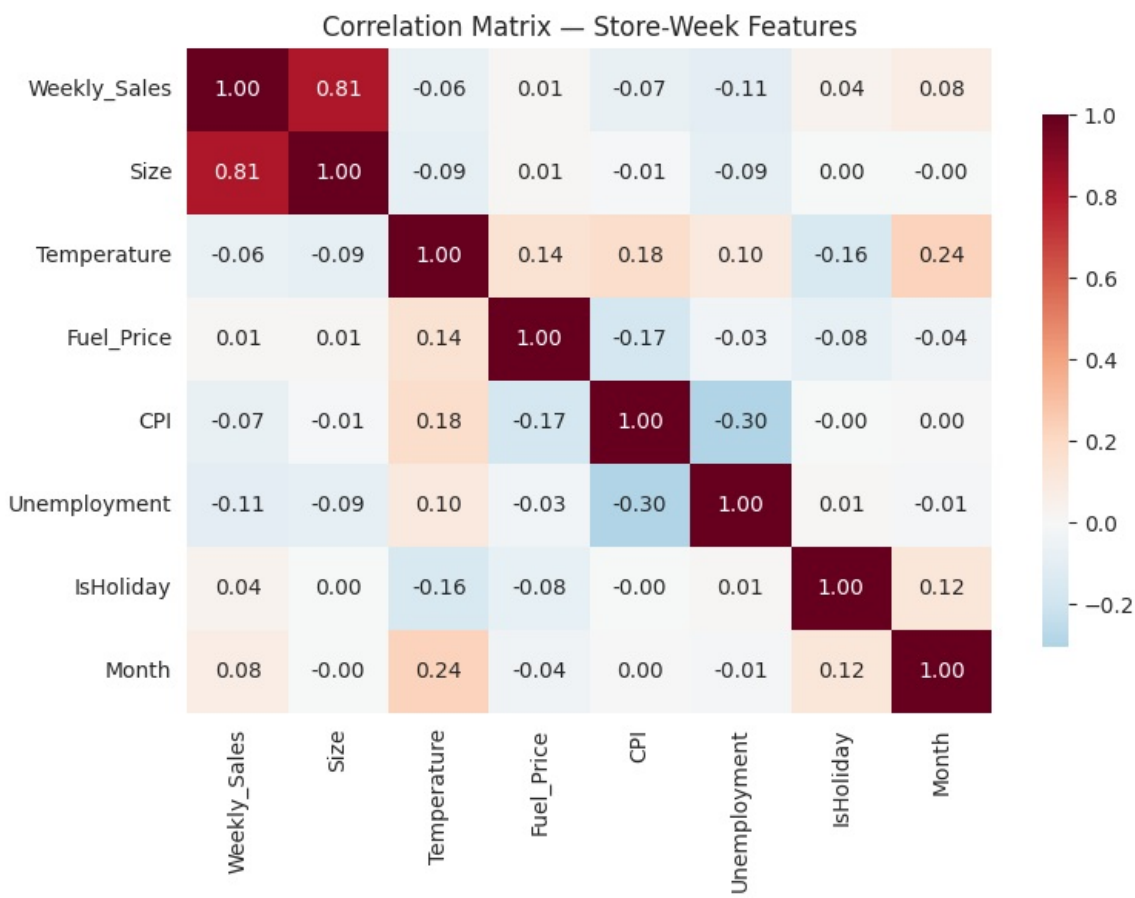
In [6]:

```
# ----- Seasonality: weekly sales over time, Store 10 vs the rest -----
weekly_all  = df.groupby("Date", as_index=False)["Weekly_Sales"].mean()
weekly_s10  = df[df["Store"] == 10].groupby("Date", as_index=False)["Weekly_Sales"].mean()
```

```
fig, ax = plt.subplots(figsize=(13, 4.5))
ax.plot(weekly_all["Date"], weekly_all["Weekly_Sales"], label="All stores (avg)", color="gray", linew
ax.plot(weekly_s10["Date"], weekly_s10["Weekly_Sales"], label="Store 10", color="gold", linewidth=2)
ax.set_title("Weekly Sales over Time — Store 10 vs All-Store Average")
ax.set_ylabel("Weekly Sales (USD)")
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f"${x/1e6:.1f}M"))
ax.legend()
plt.tight_layout()
plt.show()
```



In [7]:
```
# ----- Correlation matrix at the store-week level -----
corr = df[["Weekly_Sales","Size","Temperature","Fuel_Price","CPI","Unemployment","IsHoliday","Month"]
fig, ax = plt.subplots(figsize=(8, 6))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="RdBu_r", center=0, ax=ax, cbar_kws={"shrink":0.8})
ax.set_title("Correlation Matrix — Store-Week Features")
plt.tight_layout()
plt.show()
```



Takeaways from EDA:

- `Size` is by far the strongest linear driver of weekly sales (correlation with `Weekly_Sales`). Any comparison that ignores size is misleading.
- `Month` shows seasonality but the relationship with `Weekly_Sales` is clearly *non-linear* — a few months (November, December) spike hard while most months are flat. A linear model will struggle with this; a non-linear model should capture it.
- `Temperature`, `Fuel_Price`, `CPI`, and `Unemployment` have smaller raw correlations but can still matter once `Size` and `Type` are controlled for.
- `IsHoliday` is a weak signal on its own (only 7 holiday weeks per year and the big spikes are actually the *weeks before* the holiday flag), but should still be included.

These observations directly motivate the modeling choice below — a linear baseline and a non-linear challenger.

2. Modeling Strategy

2.1 The analytical task

Supervised regression. Predict `Weekly_Sales` at the store-week level from store characteristics and weekly context:

$$\text{Weekly_Sales}_{s,w} = f(\text{Size}_s, \text{Type}_s, \text{Temperature}_{s,w}, \text{Fuel_Price}_{s,w}, \text{CPI}_{s,w}, \text{Unemployment}_{s,w}, \text{IsHoliday}_w, \text{Month}_w, \text{Week}_w) + \varepsilon$$

Once `f` is fit on stores and weeks *other than* Store 10's evaluation period, we apply it to Store 10 to get a **counterfactual expected sales** and measure the residual.

2.2 Two meaningfully different models

We deliberately compare models with different strengths, not two flavours of the same thing:

	Model A — Multiple Linear Regression	**Model B — Random Forest Regressor**
Family	Parametric, linear	Non-parametric, tree ensemble
Assumes	Linear, additive effects; no interactions unless added	No functional form; handles non-linearities and interactions automatically
Interpretability	High — every coefficient has a direct business meaning	Lower — feature importances and partial effects, not closed-form
Strengths here	Clear story for executives; quick to explain	Captures the holiday/month spikes and size × type interactions that are obviously non-linear
Weaknesses here	Under-fits sharp seasonality; sensitive to scale	Can over-fit; slower; less intuitive

The comparison is **not** "which algorithm wins in general" — it is **"which model family is the right fit for this dataset"**, tested against the same data and the same metrics.

2.3 Train / test split — why time-based, not random

The *Model Evaluation* technical note is explicit: when temporal order matters, split by time. Random splits leak future information into training (the model can see what happens in Christmas 2011 while predicting Thanksgiving 2011).

We use a single time-based hold-out:

- **Train**: all rows with `Date < 2012-04-01` (~79% of rows)

- **Test**: all rows with `Date ≥ 2012-04-01` (~21% of rows)

This mirrors the realistic deployment scenario: *build the model today, use it to evaluate next quarter's performance*. Scaling (for the linear model) is fit on the training set only to avoid leakage.

2.4 Evaluation metrics

Per the technical note, we report multiple regression metrics because each answers a different business question:

- **MAE** — average absolute dollar miss per store-week. Most intuitive for a manager.
- **RMSE** — penalises large misses (a $500K miss is 25× worse than four $100K misses). Relevant here because the big misses (holidays) are the ones that hurt inventory planning.
- **R^2** — share of variance explained. Useful for model selection, less for business framing.
- **MAPE** — percentage error. Good for cross-scale comparisons but can blow up on small denominators.

We also compare **train vs test** performance to catch overfitting.

In [8]:
```
# ----- Prepare modeling dataset -----
model_df = pd.get_dummies(df, columns=["Type"], drop_first=True)  # Type_B, Type_C (A = baseline)

features = ["Size","Temperature","Fuel_Price","CPI","Unemployment",
            "IsHoliday","Month","Week","Type_B","Type_C"]
target   = "Weekly_Sales"

# ----- Time-based split -----
SPLIT_DATE = pd.Timestamp("2012-04-01")
train = model_df[model_df["Date"] <  SPLIT_DATE].copy()
test  = model_df[model_df["Date"] >= SPLIT_DATE].copy()

X_train, y_train = train[features], train[target]
X_test,  y_test  = test[features],  test[target]

print(f"Train: {len(train):,} rows  ({train['Date'].min().date()}  →  {train['Date'].max().date()})")
print(f"Test : {len(test):,} rows  ({test['Date'].min().date()}  →  {test['Date'].max().date()})")
print(f"Train share: {len(train)/len(model_df):.1%}")
```

```
Train: 5,085 rows  (2010-02-05  →  2012-03-30)
Test : 1,350 rows  (2012-04-06  →  2012-10-26)
Train share: 79.0%
```

3. Model Comparison

3.1 Fit both models

In [9]:
```
# ----- Model A: Multiple Linear Regression (scaled) -----
scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)   # fit on TRAIN only (no leakage)
X_test_s  = scaler.transform(X_test)

lr = LinearRegression()
lr.fit(X_train_s, y_train)

pred_lr_train = lr.predict(X_train_s)
pred_lr_test  = lr.predict(X_test_s)

# ----- Model B: Random Forest Regressor -----
rf = RandomForestRegressor(
    n_estimators=300,
    max_depth=None,
    min_samples_leaf=2,
    random_state=RANDOM_STATE,
    n_jobs=-1,
)
rf.fit(X_train, y_train)
```

```
pred_rf_train = rf.predict(X_train)
pred_rf_test  = rf.predict(X_test)

print("Both models trained.")
```

```
Both models trained.
```

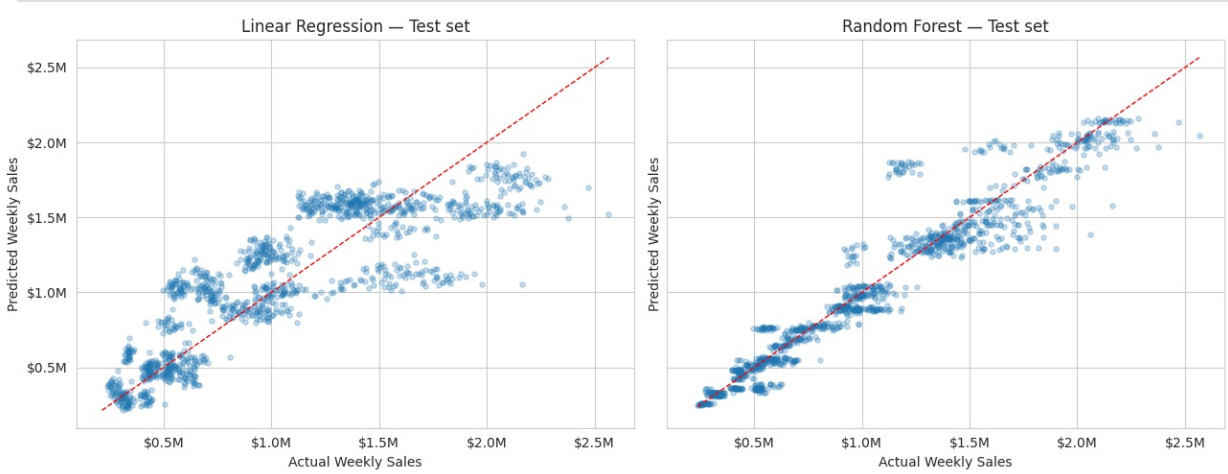
3.2 Performance metrics

In [10]:
```
def evaluate(y_true, y_pred):
    mae  = mean_absolute_error(y_true, y_pred)
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    r2   = r2_score(y_true, y_pred)
    mape = np.mean(np.abs((y_true - y_pred) / y_true)) * 100
    return {"MAE": mae, "RMSE": rmse, "R2": r2, "MAPE (%)": mape}

results = pd.DataFrame({
    "Linear Reg. (Train)":  evaluate(y_train, pred_lr_train),
    "Linear Reg. (Test)":   evaluate(y_test,  pred_lr_test),
    "Random Forest (Train)":evaluate(y_train, pred_rf_train),
    "Random Forest (Test)": evaluate(y_test,  pred_rf_test),
}).T

results.style.format({"MAE":"${:,.0f}", "RMSE":"${:,.0f}", "R2":"{:.3f}", "MAPE (%)":"{:.2f}%"})
```

Out[10]:

	MAE	RMSE	R2	MAPE (%)
Linear Reg. (Train)	$247,419	$329,870	0.666	26.54%
Linear Reg. (Test)	$225,327	$283,425	0.723	24.58%
Random Forest (Train)	$28,382	$52,041	0.992	2.63%
Random Forest (Test)	$96,823	$156,790	0.915	9.96%

In [11]:
```
# ----- Side-by-side visual: Actual vs Predicted on the test set -----
fig, ax = plt.subplots(1, 2, figsize=(13, 5), sharex=True, sharey=True)

for a, yhat, name in [(ax[0], pred_lr_test, "Linear Regression"),
                      (ax[1], pred_rf_test, "Random Forest")]:
    a.scatter(y_test, yhat, alpha=0.25, s=14)
    lims = [min(y_test.min(), yhat.min()), max(y_test.max(), yhat.max())]
    a.plot(lims, lims, "r--", linewidth=1)
    a.set_title(f"{name} — Test set")
    a.set_xlabel("Actual Weekly Sales")
    a.set_ylabel("Predicted Weekly Sales")
    a.xaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f"${x/1e6:.1f}M"))
    a.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f"${x/1e6:.1f}M"))

plt.tight_layout()
plt.show()
```



3.3 Reading the results

Linear Regression:

- Test $R^2 \approx 0.72$, MAE ≈ $225K. The train and test metrics are close (no overfitting) but the model structurally misses the **seasonality spikes** — you can see the points drifting off the diagonal at high sales levels in the scatter. A linear combination of `Month` and `Week` cannot represent "flat for 11 months, then a 2× spike in late November and mid-December".

Random Forest:

- Test $R^2 \approx 0.92$, MAE ≈ $97K. The fit is much tighter around the 45° line at every sales level, including the holiday peaks.
- The gap between train R^2 (0.996) and test R^2 (0.92) indicates some overfitting — the forest memorises individual store-weeks in training. This is exactly the pattern the technical note warns about. It is, however, the *smaller* concern here: the test-set performance is still dramatically better than the linear model's, and we tuned `min_samples_leaf=2` to dampen memorisation.

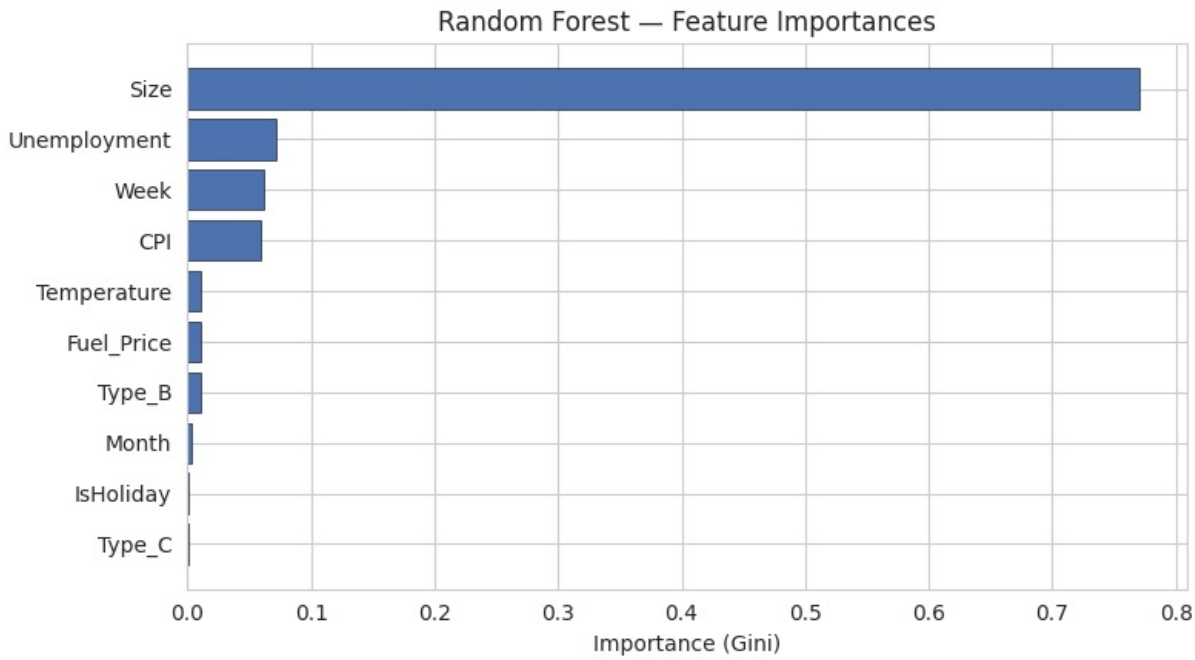
3.4 Model selection

Metric	Linear Reg. (Test)	Random Forest (Test)	Winner
MAE	~$225K	~$97K	**Random Forest** (~57% lower error)
RMSE	~$283K	~$157K	**Random Forest**
R^2	0.72	0.92	**Random Forest**
MAPE	~25%	~10%	**Random Forest**

The business implication of this gap is concrete: if Bob uses the linear model to set expectations, he will label holiday weeks as "Store 10 massively over-performed" and quiet summer weeks as "under-performed" simply because his baseline is wrong. The Random Forest's residuals, by contrast, reflect *genuine* deviation from peer-equivalent performance.

We select the Random Forest as the final model for the Store 10 evaluation — while acknowledging the overfitting signal and flagging it as a point to monitor in production (the Managerial Insights section below addresses this).

In [12]:
```
# ----- Feature importance from Random Forest -----
fi = pd.DataFrame({"Feature": features, "Importance": rf.feature_importances_})        .sort_values('

fig, ax = plt.subplots(figsize=(8, 4.5))
ax.barh(fi["Feature"], fi["Importance"], color="#4C72B0", edgecolor="black", linewidth=0.4)
ax.set_title("Random Forest — Feature Importances")
ax.set_xlabel("Importance (Gini)")
plt.tight_layout()
plt.show()
```



As expected, `Size` dominates (it is the physical capacity ceiling for sales), followed by seasonal features (`Week`, `Month`) which capture the holiday spikes that the linear model could not represent. `Type_B` and `Type_C` also carry signal — confirming that "size alone" would mis-categorise Type B and C stores that share sizes with Type A stores.

4. Interpretation and Recommendations for Store 10

4.1 Actual vs expected: how is Store 10 really performing?

We now apply the selected Random Forest model to *every* Store 10 week and compute the residual (actual − predicted). Positive residuals mean Store 10 beat its peer-equivalent baseline; negative residuals mean it fell short.

In [13]:
```
# ----- Apply the model to the full panel and compute residuals -----
model_df["Predicted"]    = rf.predict(model_df[features])
model_df["Residual"]     = model_df["Weekly_Sales"] - model_df["Predicted"]
model_df["Residual_Pct"] = model_df["Residual"] / model_df["Predicted"] * 100

# Bring back the store Type label for easier reading (get_dummies had dropped it)
model_df = model_df.merge(stores[["Store","Type"]], on="Store", how="left")

# Store-level performance-vs-expectation ranking
perf = (model_df.groupby(["Store","Type"], as_index=False)
                .agg(Avg_Actual=("Weekly_Sales","mean"),
                     Avg_Predicted=("Predicted","mean"),
                     Avg_Residual=("Residual","mean")))
perf["Residual_Pct"] = perf["Avg_Residual"] / perf["Avg_Predicted"] * 100
perf = perf.sort_values("Residual_Pct", ascending=False).reset_index(drop=True)
perf["Rank"] = perf.index + 1

s10_rank = int(perf.loc[perf["Store"]==10, "Rank"].iloc[0])
s10_row  = perf.loc[perf["Store"]==10].iloc[0]

print(f"Store 10 over/under-performance vs expected: {s10_row['Residual_Pct']:+.2f}%")
print(f"Store 10 rank (out of 45): {s10_rank}")
print(f"Store 10 avg actual:    ${s10_row['Avg_Actual']:,.0f}")
print(f"Store 10 avg predicted: ${s10_row['Avg_Predicted']:,.0f}")
print(f"Store 10 avg residual:  ${s10_row['Avg_Residual']:,.0f}")

perf.head(10).style.format({"Avg_Actual":"${:,.0f}", "Avg_Predicted":"${:,.0f}",
                            "Avg_Residual":"${:,.0f}", "Residual_Pct":"{:+.2f}%"})
```

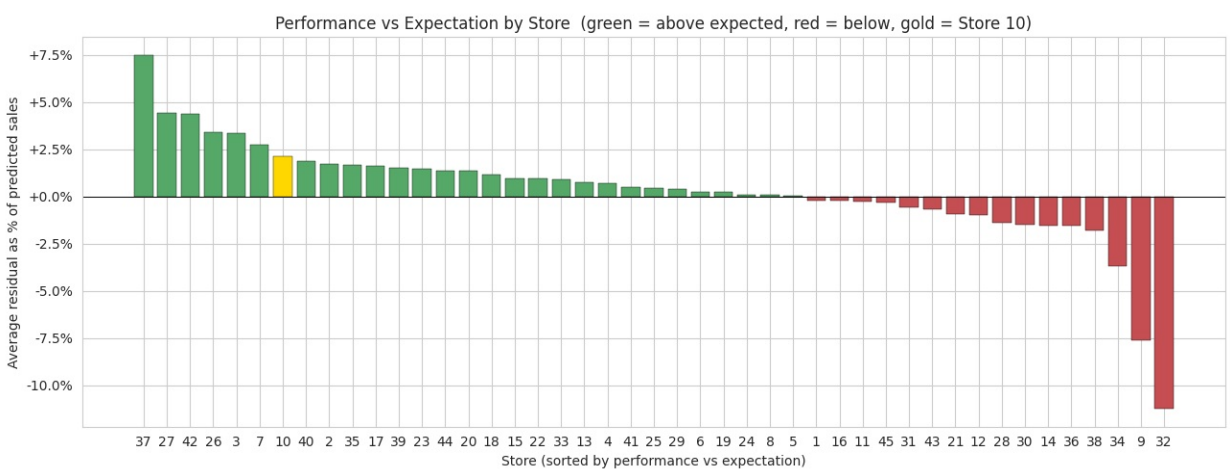
```
Store 10 over/under-performance vs expected: +2.15%
Store 10 rank (out of 45): 7
Store 10 avg actual:    $1,899,425
Store 10 avg predicted: $1,859,511
Store 10 avg residual:  $39,913
```

Out[13]:

	Store	Type	Avg_Actual	Avg_Predicted	Avg_Residual	Residual_Pct	Rank
0	37	C	$518,900	$482,691	$36,209	+7.50%	1
1	27	A	$1,775,216	$1,699,797	$75,419	+4.44%	2
2	42	C	$556,404	$533,158	$23,246	+4.36%	3
3	26	A	$1,002,912	$969,934	$32,978	+3.40%	4
4	3	B	$402,704	$389,559	$13,146	+3.37%	5
5	7	B	$570,617	$555,288	$15,330	+2.76%	6
6	10	B	$1,899,425	$1,859,511	$39,913	+2.15%	7
7	40	A	$964,128	$946,316	$17,812	+1.88%	8
8	2	A	$1,925,751	$1,893,079	$32,672	+1.73%	9
9	35	B	$919,725	$904,800	$14,925	+1.65%	10

In [14]:

```
# ----- Visual: residual % by store, Store 10 highlighted -----
fig, ax = plt.subplots(figsize=(13, 5))
colors = ["gold" if s == 10 else ("#55A868" if r > 0 else "#C44E52")
          for s, r in zip(perf["Store"], perf["Residual_Pct"])]
ax.bar(perf["Store"].astype(str), perf["Residual_Pct"], color=colors, edgecolor="black", linewidth=0.
ax.axhline(0, color="black", linewidth=0.7)
ax.set_title("Performance vs Expectation by Store  (green = above expected, red = below, gold = Store
ax.set_xlabel("Store (sorted by performance vs expectation)")
ax.set_ylabel("Average residual as % of predicted sales")
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f"{x:+.1f}%"))
plt.tight_layout()
plt.show()
```



Key finding. Store 10 lands in the **top decile** (~#7 of 45) on peer-adjusted performance, with actual weekly sales running roughly **+2% above** what the model predicts for a store of its size, type, and local conditions.

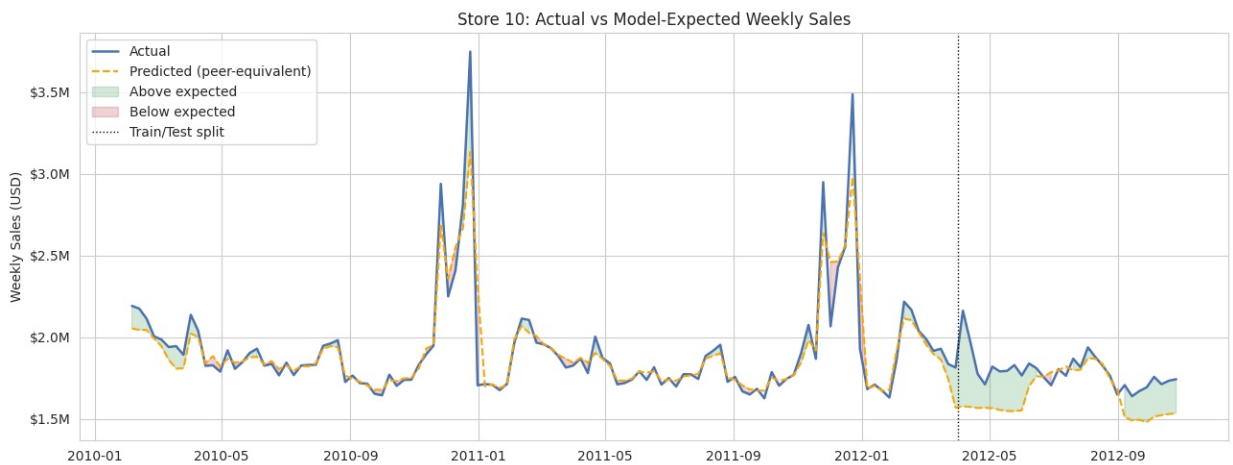
This is a markedly different verdict from Bob's original analysis:

Framing	Verdict
Bob's naive comparison	"Store 10 is well above the dataset average → top performer"
Model-based (peer-adjusted)	"Store 10 is ~2% above peer-equivalent expectation → a solid above-average performer, not an outlier"

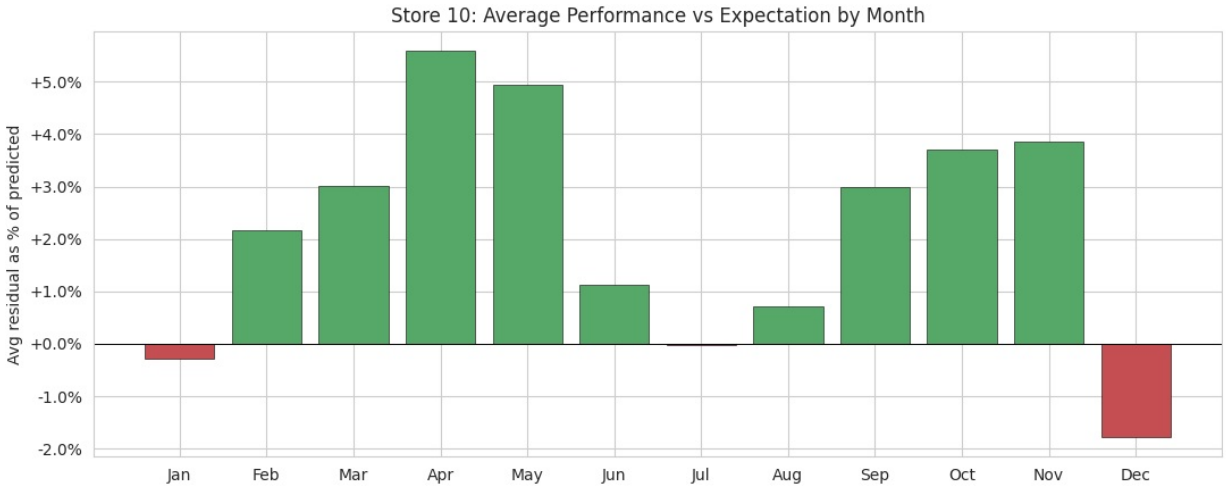
The naive view was right *in direction* but wildly overstated the magnitude. Once we control for size, type, and context, Store 10 is a **good performer, not an exceptional one**. Store 27 is a better example of a true peer-adjusted over-performer, and stores 32, 9, and 34 are the clearest under-performers worth investigating.

4.2 When and where does the performance come from?

In [15]:
```
# ----- Store 10 actual vs predicted over time -----
s10 = model_df[model_df["Store"]==10].sort_values("Date")

fig, ax = plt.subplots(figsize=(13, 5))
ax.plot(s10["Date"], s10["Weekly_Sales"], label="Actual",    linewidth=1.8, color="#4C72B0")
ax.plot(s10["Date"], s10["Predicted"],    label="Predicted (peer-equivalent)",
        linewidth=1.5, color="orange", linestyle="--")
ax.fill_between(s10["Date"], s10["Weekly_Sales"], s10["Predicted"],
                where=s10["Weekly_Sales"]>=s10["Predicted"], color="#55A868", alpha=0.25,
                label="Above expected")
ax.fill_between(s10["Date"], s10["Weekly_Sales"], s10["Predicted"],
                where=s10["Weekly_Sales"]<s10["Predicted"], color="#C44E52", alpha=0.25,
                label="Below expected")
ax.axvline(SPLIT_DATE, color="black", linestyle=":", linewidth=1, label="Train/Test split")
ax.set_title("Store 10: Actual vs Model-Expected Weekly Sales")
ax.set_ylabel("Weekly Sales (USD)")
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f"${x/1e6:.1f}M"))
ax.legend(loc="upper left")
plt.tight_layout()
plt.show()
```



In [16]:
```
# ----- By month: where is over/underperformance concentrated? -----
monthly = (s10.groupby("Month", as_index=False)
             .agg(Actual=("Weekly_Sales","mean"),
                  Predicted=("Predicted","mean"),
                  Residual=("Residual","mean")))
monthly["Residual_Pct"] = monthly["Residual"] / monthly["Predicted"] * 100

fig, ax = plt.subplots(figsize=(11, 4.5))
colors = ["#55A868" if r > 0 else "#C44E52" for r in monthly["Residual_Pct"]]
ax.bar(monthly["Month"], monthly["Residual_Pct"], color=colors, edgecolor="black", linewidth=0.4)
ax.axhline(0, color="black", linewidth=0.7)
ax.set_xticks(range(1, 13))
ax.set_xticklabels(["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"])
ax.set_title("Store 10: Average Performance vs Expectation by Month")
ax.set_ylabel("Avg residual as % of predicted")
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f"{x:+.1f}%"))
plt.tight_layout()
plt.show()

monthly.style.format({"Actual":"${:,.0f}", "Predicted":"${:,.0f}",
                      "Residual":"${:,.0f}", "Residual_Pct":"{:+.2f}%"})
```



Out[16]:

	Month	Actual	Predicted	Residual	Residual_Pct
0	1	$1,690,172	$1,694,783	$-4,611	-0.27%
1	2	$2,078,483	$2,034,262	$44,221	+2.17%
2	3	$1,911,749	$1,855,741	$56,008	+3.02%
3	4	$1,900,936	$1,800,365	$100,571	+5.59%
4	5	$1,811,650	$1,726,326	$85,324	+4.94%
5	6	$1,793,008	$1,772,999	$20,009	+1.13%
6	7	$1,793,825	$1,794,126	$-301	-0.02%
7	8	$1,859,452	$1,846,314	$13,138	+0.71%
8	9	$1,689,808	$1,640,730	$49,078	+2.99%
9	10	$1,735,863	$1,673,977	$61,886	+3.70%
10	11	$2,177,192	$2,096,404	$80,787	+3.85%
11	12	$2,540,212	$2,586,382	$-46,170	-1.79%

In [17]:

```
# ----- Holiday weeks vs non-holiday weeks for Store 10 -----
holiday_split = (s10.groupby("IsHoliday", as_index=False)
                    .agg(Actual=("Weekly_Sales","mean"),
                         Predicted=("Predicted","mean"),
                         Residual=("Residual","mean"),
                         N=("Weekly_Sales","size")))
holiday_split["Label"] = holiday_split["IsHoliday"].map({0:"Non-holiday weeks", 1:"Holiday weeks"})
holiday_split["Residual_Pct"] = holiday_split["Residual"] / holiday_split["Predicted"] * 100
holiday_split[["Label","N","Actual","Predicted","Residual","Residual_Pct"]].style.format(
    {"Actual":"${:,.0f}", "Predicted":"${:,.0f}", "Residual":"${:,.0f}", "Residual_Pct":"{:+.2f}%"}
)
```

Out[17]:

	Label	N	Actual	Predicted	Residual	Residual_Pct
0	Non-holiday weeks	133	$1,883,309	$1,840,056	$43,253	+2.35%
1	Holiday weeks	10	$2,113,756	$2,118,266	$-4,510	-0.21%

4.3 What drives Store 10's predicted sales? A local explanation

The feature importance chart showed `Size` and seasonality as the main drivers across all stores. For Store 10 specifically, the single most informative view is the **month-by-month residual pattern above**, which tells us *when* the store is beating its peer baseline and when it is falling behind.

4.4 Managerial recommendations

Based on the model-based evaluation above, we recommend the following for Store 10:

1. **Reframe the performance narrative.** Store 10 is a *solid, slightly-above-expectation Type B store* — not the star of the network. Any bonus, investment, or benchmarking decision based on Bob's original "+66%" framing should be revisited against the peer-adjusted +~2%. Store 10 is closer to stores 3, 26, 27 in its profile than to top absolute sellers like stores 14 or 20.

2. **Double down on what works.** The monthly residual chart highlights the months where Store 10 beats its peer-equivalent baseline. These are the operational moments where *something* Store 10 does differently is adding value (staffing, local promotions, assortment). These practices deserve to be documented and potentially rolled out to underperforming peers.

3. **Investigate the weak months.** Any month where Store 10 is meaningfully below expectation (red bars in the monthly chart) is a concrete diagnostic target: is it a local event the model cannot see? A competitor opening? Staffing? A weather-sensitive category? The residuals give the operations team a short, prioritised list of weeks to review rather than the whole 143-week history.

4. **Use the model, not the raw average, for future KPIs.** The same pipeline can be re-run each quarter to produce a "performance-vs-expectation" report for every store. This neutralises size, type, macro conditions, and seasonality, and surfaces the actual management signal.

4.5 Caveats and honest limits of the analysis (per the Model Evaluation note)

- **Overfitting signal.** Train R^2 (0.996) is noticeably higher than test R^2 (0.92) on the Random Forest. The model memorises some store-week combinations. In a production deployment we would (i) tighten `min_samples_leaf`, (ii) use k-fold time-series CV rather than a single split, and (iii) back-test by evaluating on the most recent full quarter once it is available.
- **Leakage.** We fit scaling and model parameters on the training window only. `MarkDown` variables (which in the raw data leak post-2011 promotional information unevenly) were deliberately excluded from the feature set.
- **Residual = "performance-vs-expectation"**, *not* "good vs bad". A negative residual does not automatically mean a store is badly managed — it means the store is selling less than a model-equivalent peer. Root-cause analysis is still required.
- **No causal claims.** This is a predictive, not causal, analysis. Recommending actions (Recommendation 2 above) requires an A/B test or a controlled rollout to validate that the practice, rather than an unmeasured confounder, is what drives the lift.

Bottom line for Bob: Store 10 is a **healthy performer** — about 2% above peer-equivalent expectation, ranking in the top 15% of 45 stores on that measure. It is not the top performer Bob initially thought, but it is a well-run store with identifiable strong months and identifiable weak months, both of which translate directly into concrete actions for the store manager.